

Assignment 4 (Cassandra)

CS-157C: NoSQL Database Systems

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3) The given JSON data contains objects where the location is a JSON array ([lon, lat]) and the zip code is stored under the key `_id`. The Cassandra schema requires `loc` to be of type `list<float>` and explicitly maps `zip`. To populate the table correctly, I wrote a Python script to read the JSON file line by line. The script extracts each object, maps the `_id` field to the `zip` column, and formats the `loc` array into a string that matches CQL's list syntax (e.g., [-72.62, 42.07]). Finally, the script writes this cleaned and reorganized data into a standard CSV file with headers, allowing it to be seamlessly imported into Cassandra using the `COPY` command.

4)

```
cqlsh:hw4> SELECT * FROM citylist LIMIT 15;
```

state	city	zip	loc	pop
IA	ACKLEY	50601	[-93.06084, 42.55272]	2316
IA	ACKWORTH	50001	[-93.37672, 41.37372]	491
IA	ADAIR	50002	[-94.64436, 41.51369]	1748
IA	ADEL	50003	[-94.03796, 41.62214]	4884
IA	AFTON	50830	[-94.19405, 41.04015]	1450
IA	AGENCY	52530	[-92.31623, 40.99569]	1233
IA	AINSWORTH	52201	[-91.54717, 41.32022]	1265
IA	AKRON	51001	[-96.52245, 42.83539]	2068
IA	ALBERT CITY	50510	[-94.98238, 42.7784]	1293
IA	ALBIA	52531	[-92.79462, 41.02872]	6397
IA	ALBION	50005	[-92.98821, 42.11427]	773
IA	ALBURNETT	52202	[-91.63924, 42.15832]	1038
IA	ALDEN	50006	[-93.38412, 42.51789]	1374
IA	ALEXANDER	50420	[-93.445, 42.81139]	664
IA	ALGONA	50511	[-94.23064, 43.06598]	7829

(15 rows)

```
cqlsh:hw4>
```

5)

Method 1

```
cqlsh:hw4>
cqlsh:hw4> SELECT * FROM citylist WHERE city = 'NEW YORK' LIMIT 15 ALLOW
FILTERING;
```

state	city	zip	loc	pop
NY	NEW YORK	10001	[-73.9967, 40.74838]	18913
NY	NEW YORK	10002	[-73.98768, 40.71523]	84143
NY	NEW YORK	10003	[-73.98922, 40.73125]	51224
NY	NEW YORK	10005	[-74.00835, 40.70565]	202
NY	NEW YORK	10006	[-74.01347, 40.70845]	119
NY	NEW YORK	10007	[-74.00702, 40.71391]	3374
NY	NEW YORK	10009	[-73.97959, 40.72619]	57426
NY	NEW YORK	10010	[-73.98133, 40.73748]	24907
NY	NEW YORK	10011	[-73.99963, 40.74023]	46560
NY	NEW YORK	10012	[-73.99828, 40.72553]	26365
NY	NEW YORK	10013	[-74.00253, 40.71851]	21860
NY	NEW YORK	10014	[-74.00542, 40.73393]	31147
NY	NEW YORK	10016	[-73.97813, 40.74428]	51561
NY	NEW YORK	10017	[-73.97066, 40.75172]	12465
NY	NEW YORK	10018	[-73.9925, 40.75471]	4834

(15 rows)

Tracing session: df17ce70-3e93-11f1-bc5e-a11212f30b26

activity
timestamp source source_elapsed client
-----+-----
---+-----+-----+-----

Execute CQL3 query | 2026-04-22 21:40:24.921000 | 172.17.0.2 |
0 | 127.0.0.1

Parsing SELECT * FROM citylist WHERE city = 'NEW YORK' LIMIT
15 ALLOW FILTERING; [Native-Transport-Requests-1] | 2026-04-22
21:40:24.931000 | 172.17.0.2 | 10937 | 127.0.0.1

Preparing statement [Native-Transport-Requests-1] | 2026-04-22
21:40:24.938000 | 172.17.0.2 | 18438 | 127.0.0.1

```

Computing ranges to query [Native-Transport-Requests-1] | 2026-04-22
21:40:24.947000 | 172.17.0.2 |          27043 | 127.0.0.1
  Submitting range requests on 17 ranges with a concurrency of 1 (0.0 rows
per range expected) [Native-Transport-Requests-1] | 2026-04-22
21:40:24.947001 | 172.17.0.2 |          27641 | 127.0.0.1
                                           Submitted 1
concurrent range requests [Native-Transport-Requests-1] | 2026-04-22
21:40:24.950000 | 172.17.0.2 |          29902 | 127.0.0.1
      Executing seq scan across 0 sstables for
(min(-9223372036854775808), min(-9223372036854775808)) [ReadStage-2] |
2026-04-22 21:40:24.950001 | 172.17.0.2 |          30108 | 127.0.0.1
                                                    Read
15 live rows and 0 tombstone cells [ReadStage-2] | 2026-04-22
21:40:24.964000 | 172.17.0.2 |          44391 | 127.0.0.1

Request complete | 2026-04-22 21:40:24.970906 | 172.17.0.2 |          49906
| 127.0.0.1

cqlsh:hw4>

```

Execution Time: ~49.9 ms

Method 2

CQL command to create an index:

```
CREATE INDEX city_idx ON citylist (city);
```

CQL query to get the result:

```

cqlsh:hw4>
cqlsh:hw4> SELECT * FROM citylist WHERE city = 'NEW YORK' LIMIT 15;

```

state	city	zip	loc	pop
NY	NEW YORK	10001	[-73.9967, 40.74838]	18913
NY	NEW YORK	10002	[-73.98768, 40.71523]	84143
NY	NEW YORK	10003	[-73.98922, 40.73125]	51224
NY	NEW YORK	10005	[-74.00835, 40.70565]	202
NY	NEW YORK	10006	[-74.01347, 40.70845]	119

NY	NEW YORK	10007	[-74.00702, 40.71391]	3374
NY	NEW YORK	10009	[-73.97959, 40.72619]	57426
NY	NEW YORK	10010	[-73.98133, 40.73748]	24907
NY	NEW YORK	10011	[-73.99963, 40.74023]	46560
NY	NEW YORK	10012	[-73.99828, 40.72553]	26365
NY	NEW YORK	10013	[-74.00253, 40.71851]	21860
NY	NEW YORK	10014	[-74.00542, 40.73393]	31147
NY	NEW YORK	10016	[-73.97813, 40.74428]	51561
NY	NEW YORK	10017	[-73.97066, 40.75172]	12465
NY	NEW YORK	10018	[-73.9925, 40.75471]	4834

(15 rows)

Tracing session: 2e21e050-3e94-11f1-bc5e-a11212f30b26

activity	timestamp	source	source_elapsed	client

-----+-----				
-----+-----+-----+-----				

Execute CQL3 query | 2026-04-22 21:42:37.526000 | 172.17.0.2 |
0 | 127.0.0.1

Parsing SELECT * FROM citylist WHERE city
= 'NEW YORK' LIMIT 15; [Native-Transport-Requests-1] | 2026-04-22
21:42:37.530000 | 172.17.0.2 | 4918 | 127.0.0.1

Preparing statement [Native-Transport-Requests-1] | 2026-04-22
21:42:37.533000 | 172.17.0.2 | 7073 | 127.0.0.1

Index mean cardinalities are city_idx:1.
Scanning with city_idx. [Native-Transport-Requests-1] | 2026-04-22
21:42:37.547000 | 172.17.0.2 | 21526 | 127.0.0.1

Computing ranges to query [Native-Transport-Requests-1] | 2026-04-22
21:42:37.548000 | 172.17.0.2 | 22125 | 127.0.0.1

Submitting range requests on 17 ranges with a concurrency of 17 (0.05625
rows per range expected) [Native-Transport-Requests-1] | 2026-04-22
21:42:37.548001 | 172.17.0.2 | 22395 | 127.0.0.1

Submitted 1
concurrent range requests [Native-Transport-Requests-1] | 2026-04-22
21:42:37.549000 | 172.17.0.2 | 23924 | 127.0.0.1

Executing
read on hw4.citylist using index city_idx [ReadStage-2] | 2026-04-22

```
21:42:37.554000 | 172.17.0.2 |          28394 | 127.0.0.1
                                                    Executing
single-partition query on citylist.city_idx [ReadStage-2] | 2026-04-22
21:42:37.554001 | 172.17.0.2 |          28626 | 127.0.0.1

Acquiring sstable references [ReadStage-2] | 2026-04-22 21:42:37.554002 |
172.17.0.2 |          28664 | 127.0.0.1
                                                    Skipped 0/1 non-slice-intersecting
sstables, included 0 due to tombstones [ReadStage-2] | 2026-04-22
21:42:37.554003 | 172.17.0.2 |          28871 | 127.0.0.1

Partition index found for sstable 1, size = 0 [ReadStage-2] | 2026-04-22
21:42:37.555000 | 172.17.0.2 |          29972 | 127.0.0.1

Executing single-partition query on citylist [ReadStage-2] | 2026-04-22
21:42:37.561000 | 172.17.0.2 |          35419 | 127.0.0.1

Acquiring sstable references [ReadStage-2] | 2026-04-22 21:42:37.561001 |
172.17.0.2 |          35531 | 127.0.0.1
                                                    Skipped 0/1 non-slice-intersecting
sstables, included 0 due to tombstones [ReadStage-2] | 2026-04-22
21:42:37.561002 | 172.17.0.2 |          35774 | 127.0.0.1

Key cache hit for sstable 1, size = 2 [ReadStage-2] | 2026-04-22
21:42:37.561003 | 172.17.0.2 |          35873 | 127.0.0.1

Merged data from memtables and 1 sstables [ReadStage-2] | 2026-04-22
21:42:37.563000 | 172.17.0.2 |          37780 | 127.0.0.1

Read 15 live rows and 0 tombstone cells [ReadStage-2] | 2026-04-22
21:42:37.564000 | 172.17.0.2 |          38803 | 127.0.0.1

Merged data from memtables and 1 sstables [ReadStage-2] | 2026-04-22
21:42:37.564001 | 172.17.0.2 |          38863 | 127.0.0.1

Request complete | 2026-04-22 21:42:37.570542 | 172.17.0.2 |          44542
| 127.0.0.1

cqlsh:hw4>
```

Execution Time: ~44.5 ms

Method 3

CQL command to create a materialized view:

```
CREATE MATERIALIZED VIEW citylist_mv AS
  SELECT * FROM citylist
  WHERE city IS NOT NULL AND state IS NOT NULL AND zip IS NOT NULL
  PRIMARY KEY (city, state, zip);
```

CQL query to get the result:

```
cqlsh:hw4>
cqlsh:hw4> SELECT * FROM citylist_mv WHERE city = 'NEW YORK' LIMIT 15;
```

city	state	zip	loc	pop
NEW YORK	NY	10001	[-73.9967, 40.74838]	18913
NEW YORK	NY	10002	[-73.98768, 40.71523]	84143
NEW YORK	NY	10003	[-73.98922, 40.73125]	51224
NEW YORK	NY	10005	[-74.00835, 40.70565]	202
NEW YORK	NY	10006	[-74.01347, 40.70845]	119
NEW YORK	NY	10007	[-74.00702, 40.71391]	3374
NEW YORK	NY	10009	[-73.97959, 40.72619]	57426
NEW YORK	NY	10010	[-73.98133, 40.73748]	24907
NEW YORK	NY	10011	[-73.99963, 40.74023]	46560
NEW YORK	NY	10012	[-73.99828, 40.72553]	26365
NEW YORK	NY	10013	[-74.00253, 40.71851]	21860
NEW YORK	NY	10014	[-74.00542, 40.73393]	31147
NEW YORK	NY	10016	[-73.97813, 40.74428]	51561
NEW YORK	NY	10017	[-73.97066, 40.75172]	12465
NEW YORK	NY	10018	[-73.9925, 40.75471]	4834

(15 rows)

Tracing session: 368af950-3e96-11f1-96a4-ab665de7ab24

activity	timestamp	source	source_elapsed	client

Execute CQL3 query | 2026-04-22 21:57:10.650000 | 172.17.0.2 |
0 | 127.0.0.1

```

Parsing SELECT * FROM citylist_mv WHERE city = 'NEW YORK' LIMIT 15;
[Native-Transport-Requests-1] | 2026-04-22 21:57:10.670000 | 172.17.0.2 |
23391 | 127.0.0.1

                                Preparing statement
[Native-Transport-Requests-1] | 2026-04-22 21:57:10.673000 | 172.17.0.2 |
26561 | 127.0.0.1

                                Executing single-partition query on
citylist_mv [ReadStage-2] | 2026-04-22 21:57:10.683000 | 172.17.0.2 |
36915 | 127.0.0.1

                                Acquiring sstable
references [ReadStage-2] | 2026-04-22 21:57:10.683001 | 172.17.0.2 |
37096 | 127.0.0.1

                                Skipped 0/0 non-slice-intersecting sstables, included 0 due to
tombstones [ReadStage-2] | 2026-04-22 21:57:10.684000 | 172.17.0.2 |
37911 | 127.0.0.1

                                Merged data from memtables and 0
sstables [ReadStage-2] | 2026-04-22 21:57:10.686000 | 172.17.0.2 |
40185 | 127.0.0.1

                                Read 15 live rows and 0
tombstone cells [ReadStage-2] | 2026-04-22 21:57:10.687000 | 172.17.0.2 |
40508 | 127.0.0.1

Request complete | 2026-04-22 21:57:10.692041 | 172.17.0.2 |           42041
| 127.0.0.1

cqlsh:hw4>

```

Execution Time: ~42.0 ms

- **Method 1:** ALLOW FILTERING scans all partitions across the entire dataset, loading every single row to check the condition. This usually results in the longest execution time and is very inefficient for large datasets.
 - **Method 2:** A Secondary Index creates a hidden index table. Cassandra first queries the index to find matching primary keys, and then fetches those rows from the base table. Execution time is faster than Method 1, but there is still overhead due to looking up non-contiguous data.
 - **Method 3:** A Materialized View actively maintains a copy of the data reorganized using city as the partition key. Fetching city = 'NEW YORK' is an extremely fast exact partition key lookup, making it the most efficient of the three methods.
-

6)

Method 3 (Materialized View) combined with ALLOW FILTERING works perfectly because the Materialized View's partition key is city. Method 2 (Secondary Index) with ALLOW FILTERING also works. However, using the Materialized View is considered much better practice.

```
cqlsh:hw4>
cqlsh:hw4> SELECT zip FROM citylist_mv WHERE city = 'NEW YORK' AND pop >
20000 AND pop < 30000 LIMIT 15 ALLOW FILTERING;
```

```
zip
-----
10010
10012
10013
10026
10030
10035
10039
```

(7 rows)

Tracing session: 73aadcb0-3e96-11f1-96a4-ab665de7ab24

```
activity
| timestamp                | source      | source_elapsed | client
-----+-----+-----+-----+
-----+-----+-----+-----+
```

```
Execute CQL3 query | 2026-04-22 21:58:53.179000 | 172.17.0.2 |
0 | 127.0.0.1
```

```
Parsing SELECT zip FROM citylist_mv WHERE city = 'NEW YORK' AND pop >
20000 AND pop < 30000 LIMIT 15 ALLOW FILTERING;
[Native-Transport-Requests-1] | 2026-04-22 21:58:53.184000 | 172.17.0.2 |
4742 | 127.0.0.1
```

```
Preparing statement [Native-Transport-Requests-1] | 2026-04-22
21:58:53.201000 | 172.17.0.2 |          22134 | 127.0.0.1
```

```
Executing single-partition query on citylist_mv [ReadStage-2] | 2026-04-22
21:58:53.221000 | 172.17.0.2 |          42290 | 127.0.0.1
```

```
Acquiring sstable references [ReadStage-2] | 2026-04-22 21:58:53.221001 |
```

```

172.17.0.2 |          42505 | 127.0.0.1
                                                    Skipped 0/0
non-slice-intersecting sstables, included 0 due to tombstones [ReadStage-2]
| 2026-04-22 21:58:53.222000 | 172.17.0.2 |          43019 | 127.0.0.1

Merged data from memtables and 0 sstables [ReadStage-2] | 2026-04-22
21:58:53.224000 | 172.17.0.2 |          45519 | 127.0.0.1

Read 40 live rows and 0 tombstone cells [ReadStage-2] | 2026-04-22
21:58:53.225000 | 172.17.0.2 |          45787 | 127.0.0.1

Request complete | 2026-04-22 21:58:53.225727 | 172.17.0.2 |          46727
| 127.0.0.1

cqlsh:hw4>

```

7)

```

CREATE TABLE citylist2 ( city varchar, loc list<float>, pop int, state
varchar, zip varchar, PRIMARY KEY (state, city, zip) ) WITH CLUSTERING
ORDER BY (city ASC, zip DESC);

```

8)

```

cqlsh:hw4>
cqlsh:hw4> SELECT * FROM citylist2 LIMIT 15;

```

state	city	zip	loc	pop
IA	ACKLEY	50601	[-93.06084, 42.55272]	2316
IA	ACKWORTH	50001	[-93.37672, 41.37372]	491
IA	ADAIR	50002	[-94.64436, 41.51369]	1748
IA	ADEL	50003	[-94.03796, 41.62214]	4884
IA	AFTON	50830	[-94.19405, 41.04015]	1450
IA	AGENCY	52530	[-92.31623, 40.99569]	1233
IA	AINSWORTH	52201	[-91.54717, 41.32022]	1265
IA	AKRON	51001	[-96.52245, 42.83539]	2068
IA	ALBERT CITY	50510	[-94.98238, 42.7784]	1293
IA	ALBIA	52531	[-92.79462, 41.02872]	6397
IA	ALBION	50005	[-92.98821, 42.11427]	773

```
IA | ALBURNETT | 52202 | [-91.63924, 42.15832] | 1038
IA | ALDEN | 50006 | [-93.38412, 42.51789] | 1374
IA | ALEXANDER | 50420 | [-93.445, 42.81139] | 664
IA | ALGONA | 50511 | [-94.23064, 43.06598] | 7829
```

```
(15 rows)
cqlsh:hw4>
```

9)

```
cqlsh:hw4>
cqlsh:hw4> SELECT zip FROM citylist2 WHERE state = 'CA' LIMIT 15;
```

```
zip
-----
95220
93510
93446
92301
96006
91350
92536
93601
94507
94706
95410
91803
91801
92656
95910
```

```
(15 rows)
cqlsh:hw4>
```

10)

```
cqlsh:hw4>
cqlsh:hw4> SELECT zip FROM citylist2 WHERE state = 'CA' AND city = 'SAN
JOSE' AND pop > 5000 AND pop < 10000 LIMIT 15 ALLOW FILTERING;
```

```
zip
```

```
-----
```

```
95139
```

```
95138
```

```
95135
```

```
95119
```

```
(4 rows)
```

```
cqlsh:hw4>
```
